

How it works

This project uses two main turtles: one to draw the caterpillar and one to draw the leaves. The code places each new leaf at a random location. When the program detects that a leaf has been eaten, the variables storing the score, the speed of the caterpillar, and its length are increased. A function figures out if the caterpillar has moved outside the window, which would signal the end of the game.

◀ Caterpillar flowchart

To make the caterpillar move across the screen you'll use an infinite loop. Each time the loop goes round, the caterpillar moves forwards slightly. When the loop repeats quickly, these small movements create the illusion that your caterpillar is crawling.

First steps

For such a fun game, the code is surprisingly straightforward. You'll start by setting up the turtles, before moving on to the main game loop and finally the keyboard controls.

1 Getting started

Open IDLE and create a new file. Save it as "caterpillar.py".

2 Import the modules

Add these two `import` statements to tell Python that you need the `turtle` and `random` modules. The third line sets the background colour for the game window.

```
import random
import turtle as t

t.bgcolor('yellow')
```

This adds a yellow background.

